Course: CS213 Object Oriented Programming (2026/2027 First Semester)

Faculty: Faculty of Computers and Artificial Intelligence, Cairo University

Instructor: Dr. Mohammad El-Ramly

Assignment 2 part 2

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Work breakdown

Andrew Fawzy Mina

Video , PDF , 3rd Design , Classes : (VoleMachine, SimulatorUI)

Samy Atef Masoud

Program 2, 1st AI Design, Classes: (IstructionDecoder, Instruction, LoadInstruction, StoreIsntruction)

Ali Sameh Mohamed

Final and Initial Designs, Classes : (MoveInstruction, AthematicInstruction , JumpInstruction , HaltInstruction)

Omar Ashraf Mostafa Helal

Program 1 , 2nd design, Classes : (Memory, RegisterFile)

introduction

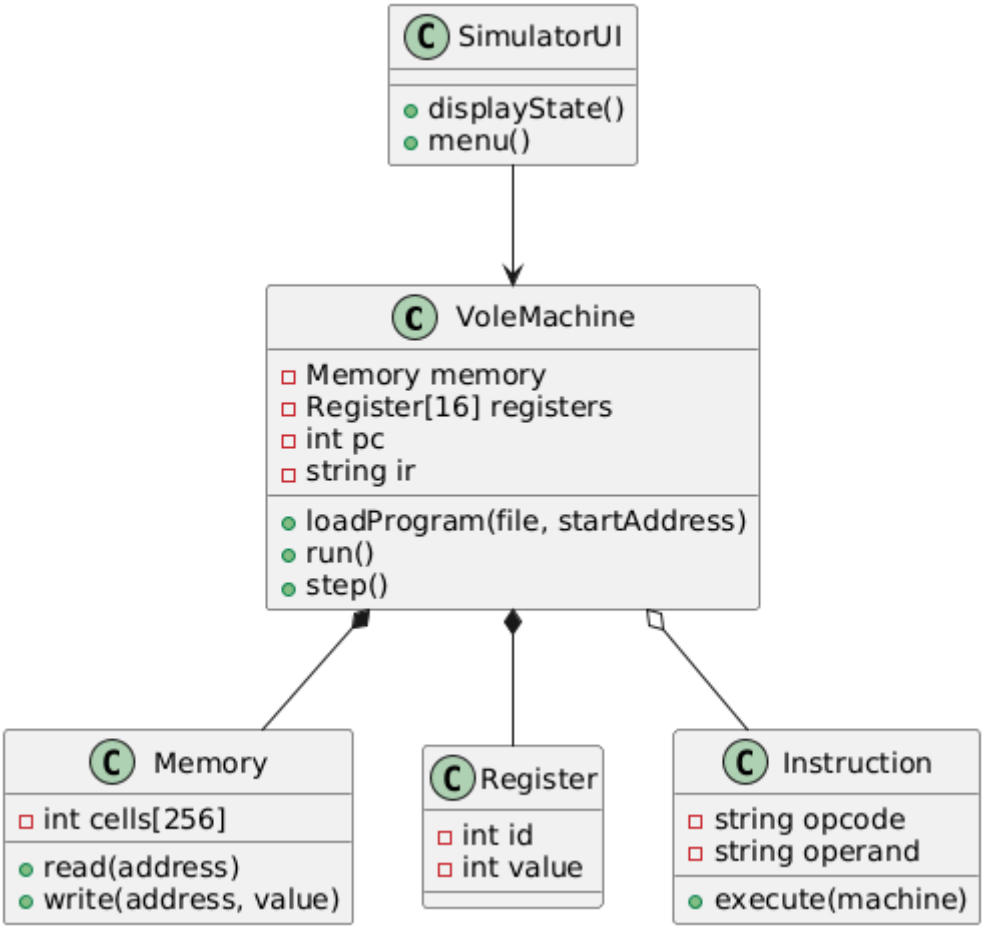
The **Vole Machine** is an educational, simplified virtual computer designed to help students understand foundational computer architecture and low-level execution mechanics. Rather than utilizing complex modern hardware architectures, the Vole machine abstracts computer components into a streamlined model consisting of:

- **16 General-Purpose Registers** for fast data manipulation.
- **256 Bytes of Main Memory** for storing instructions and data.
- **A Central Processing Unit (CPU)** responsible for the fetch-decode-execute cycle, handling arithmetic operations, and managing program flow.

By sequentially processing short instruction sets—loading values, performing arithmetic, and committing results—the simulator provides a transparent environment for learning object-oriented software design principles alongside computer organization fundamentals.

Initial Design

This PlantUML diagrams show how the code is organized into different parts and how the system design improves step by step. The design connects a user menu (SimulatorUI) to the main computer (VoleMachine), which uses memory, registers, and specific instruction classes to run programs easily.



Classes and their responsibilities :

-SimulatorUI:

Acts as the user-facing layer, connecting user menu choices to the main computer and presenting the current status/state of the simulator.

-VoleMachine

Serves as the core central processing unit (CPU) model. It manages main memory, the 16 general-purpose registers, tracks execution flow via the program counter and instruction register, and handles loading and executing programs.

-Memory

Represents the 256 bytes of main memory used for storing instructions and data, providing access control to read from or write to specific memory addresses.

-Register

Represents individual general-purpose registers used for fast data manipulation within the machine.

-Instruction

Represents the instruction set components, parsing the opcode and operand to perform specific machine operations when executed.

1st AI Design

Conversation link: https://copilot.microsoft.com/shares/XScXNMF9b33s2SvRheksT

Key Improvements:

- Introduced an InstructionDecoder component to decouple parsing logic from the execution flow.
- Grouped registers into a dedicated RegisterFile class to encapsulate register management.
- Added validation methods and state-dump utilities for enhanced debugging.

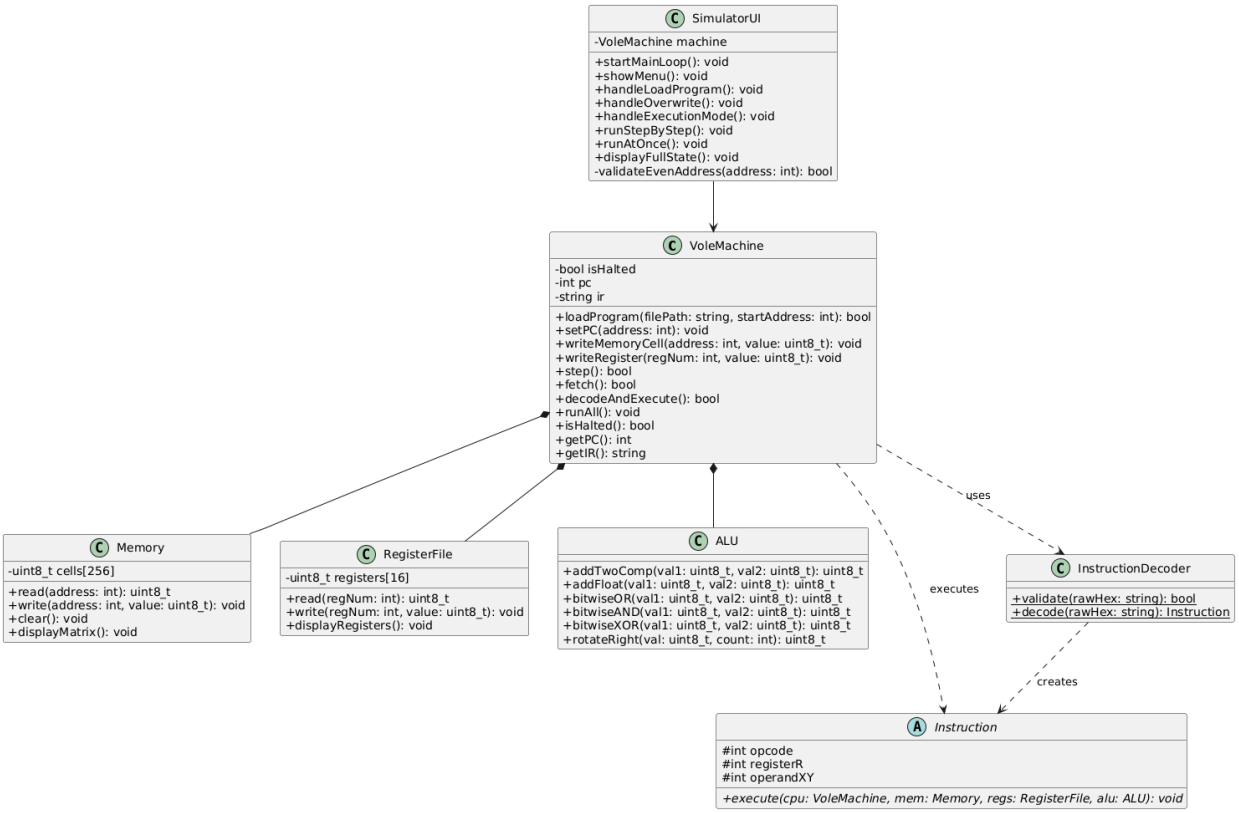


2nd AI Design

Conversation link: https://share.gemini.google/IViSnYbCynPZ

Key Improvements:

- Integrated an ALU (Arithmetic Logic Unit) class to explicitly separate computational logic (two's complement addition, floating-point math, and bitwise operations) from control flow.
- Standardized data types (utilizing fixed-width integer types such as uint8_t).
- Made Instruction an abstract base class to support specialized operation sub-typing.

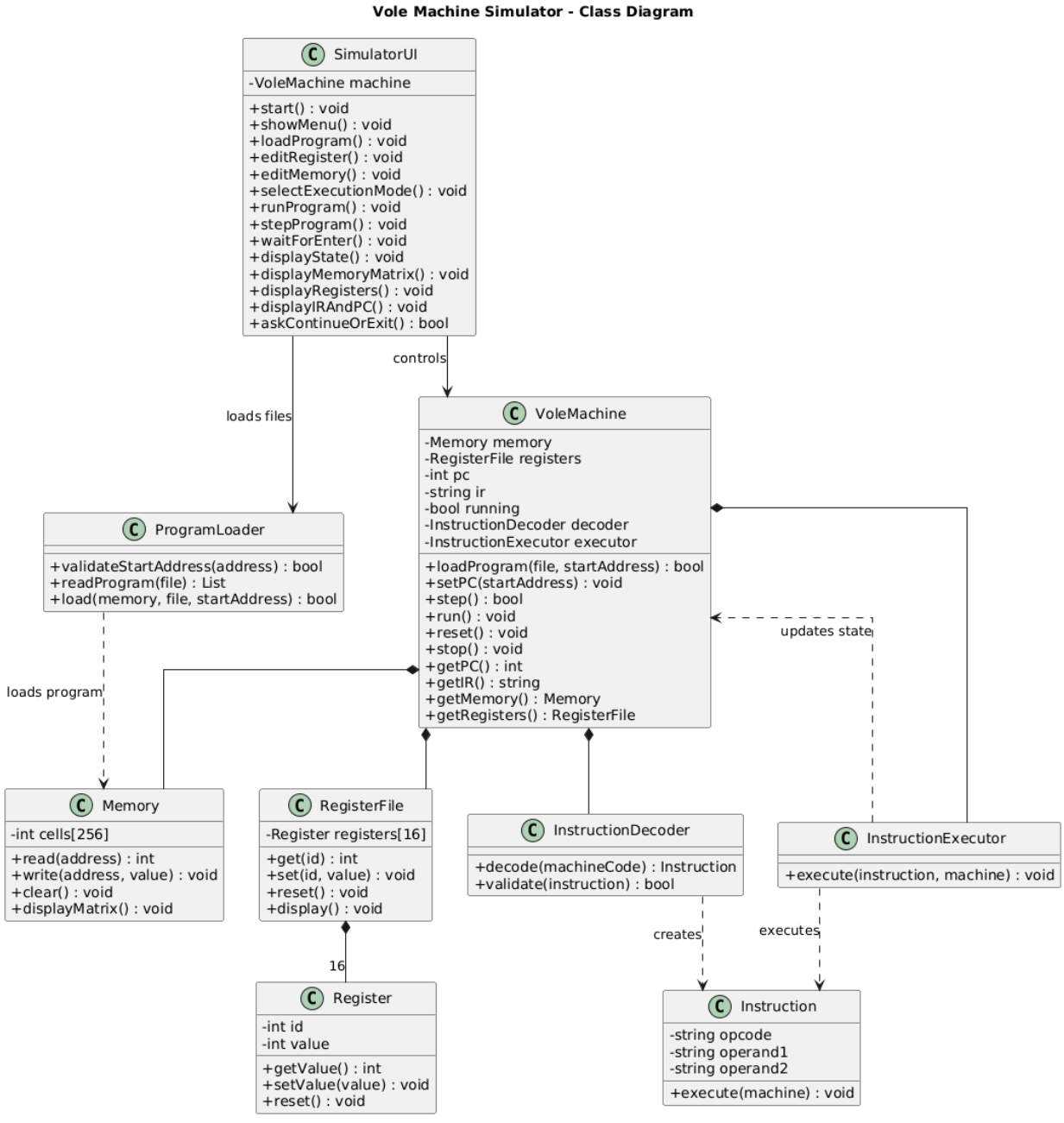


3rd AI Design

Conversation link: https://chatgpt.com/share/6add0fbb-82a8-83ea-b15f-b3cd9cb02250

Key Improvements:

- Separated file parsing concerns into a dedicated ProgramLoader class to follow the Single Responsibility Principle (SRP).
- Added an InstructionExecutor component to delegate execution paths cleanly away from the central machine state.

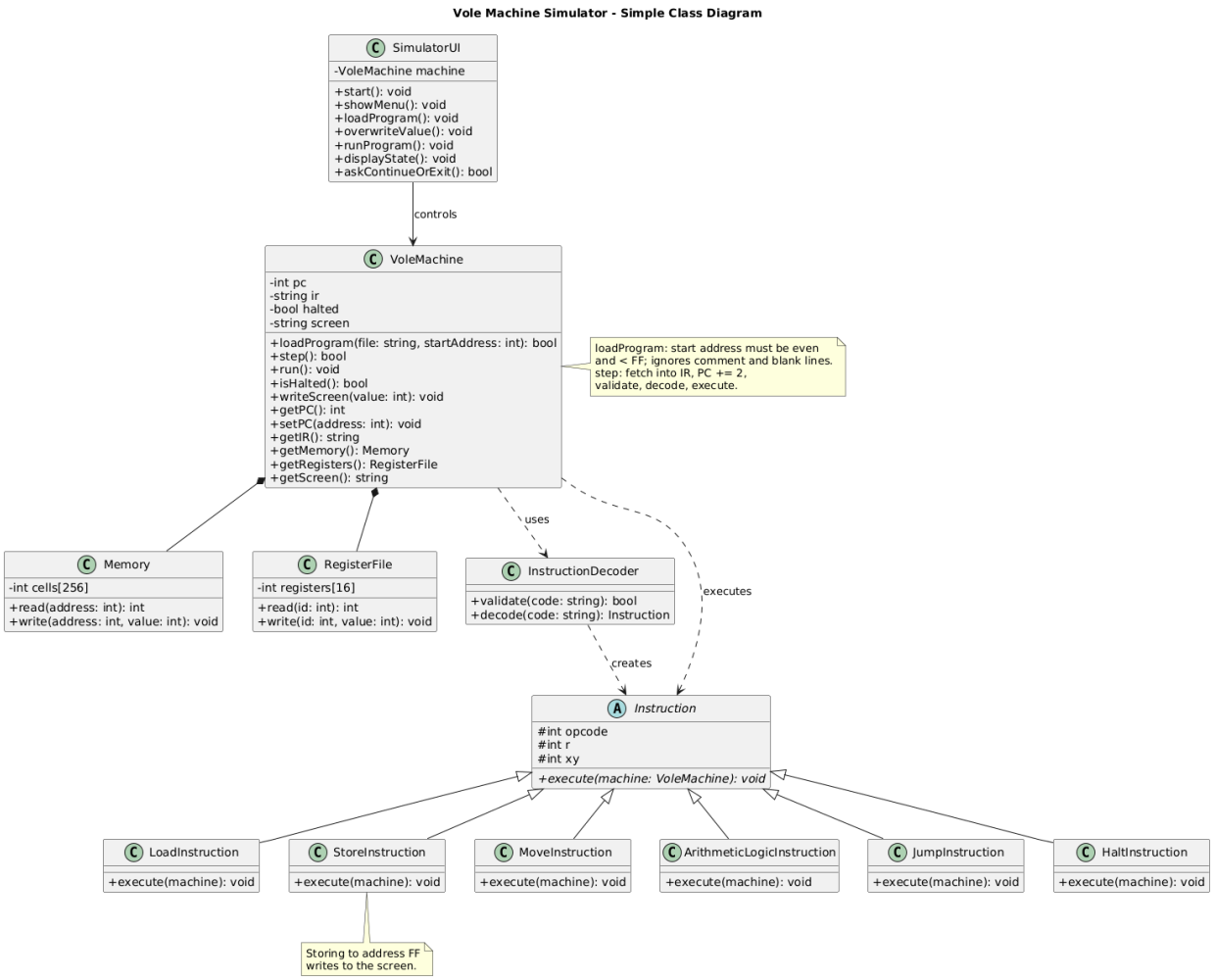


Final Design

Conversation link: https://claude.ai/share/2f7a249d-f69d-4177-bc1f-6f50c31e9bdd

Key Improvements:

- Consolidated and streamlined the architecture for optimal maintainability and clean C++ implementation.
- Implemented robust polymorphic instruction subclasses (LoadInstruction, StoreInstruction, MoveInstruction, ArithmeticLogicInstruction, JumpInstruction, and HaltInstruction) extending an abstract Instruction base class.
- Integrated screen/output handling directly into the system state and refined parser/execution notes for edge cases (e.g., memory bounds and text formatting).



Examples

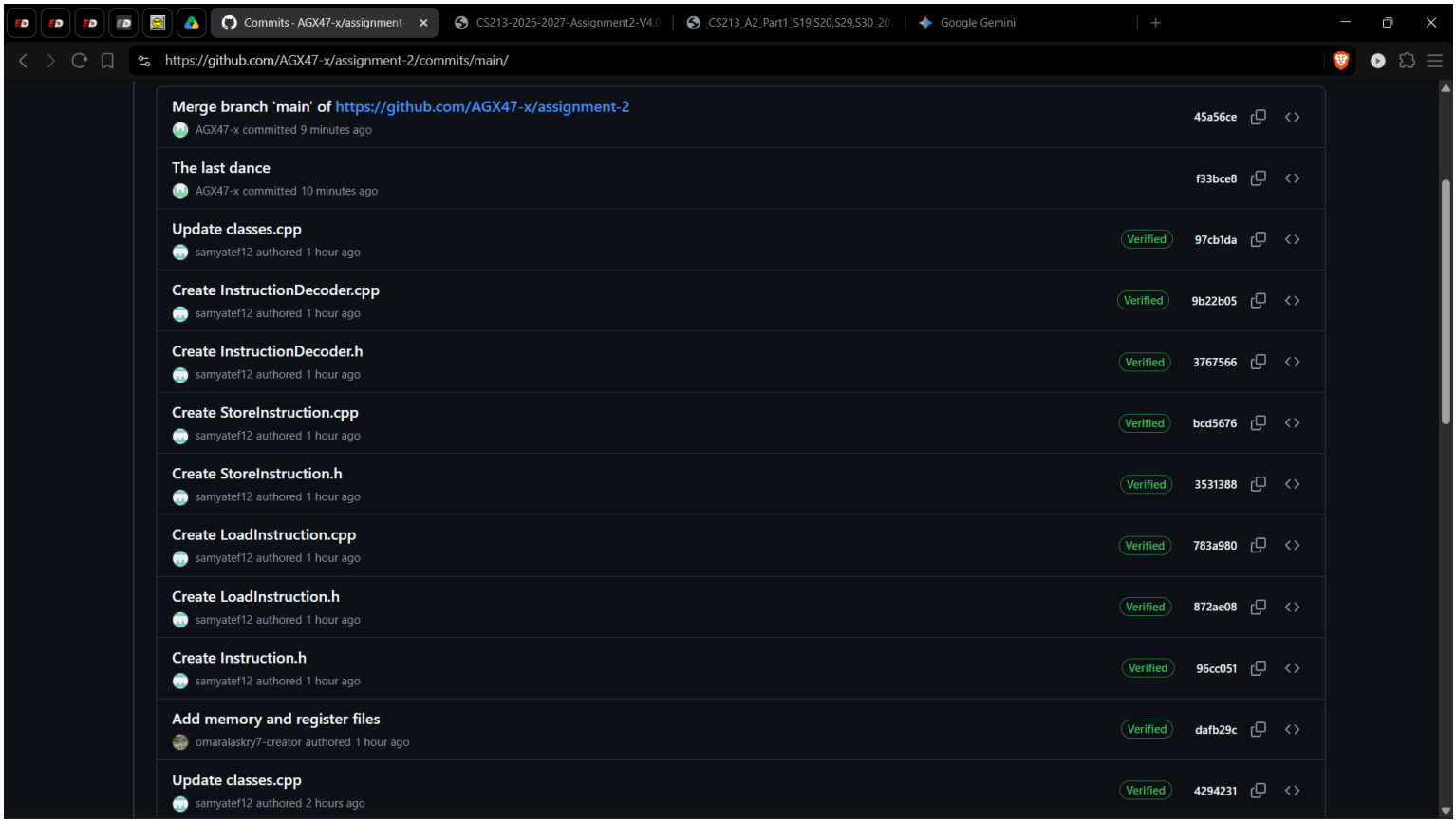
; Program 1: Addition and Bitwise AND
; This program adds three numbers and applies a Bitwise AND mask.

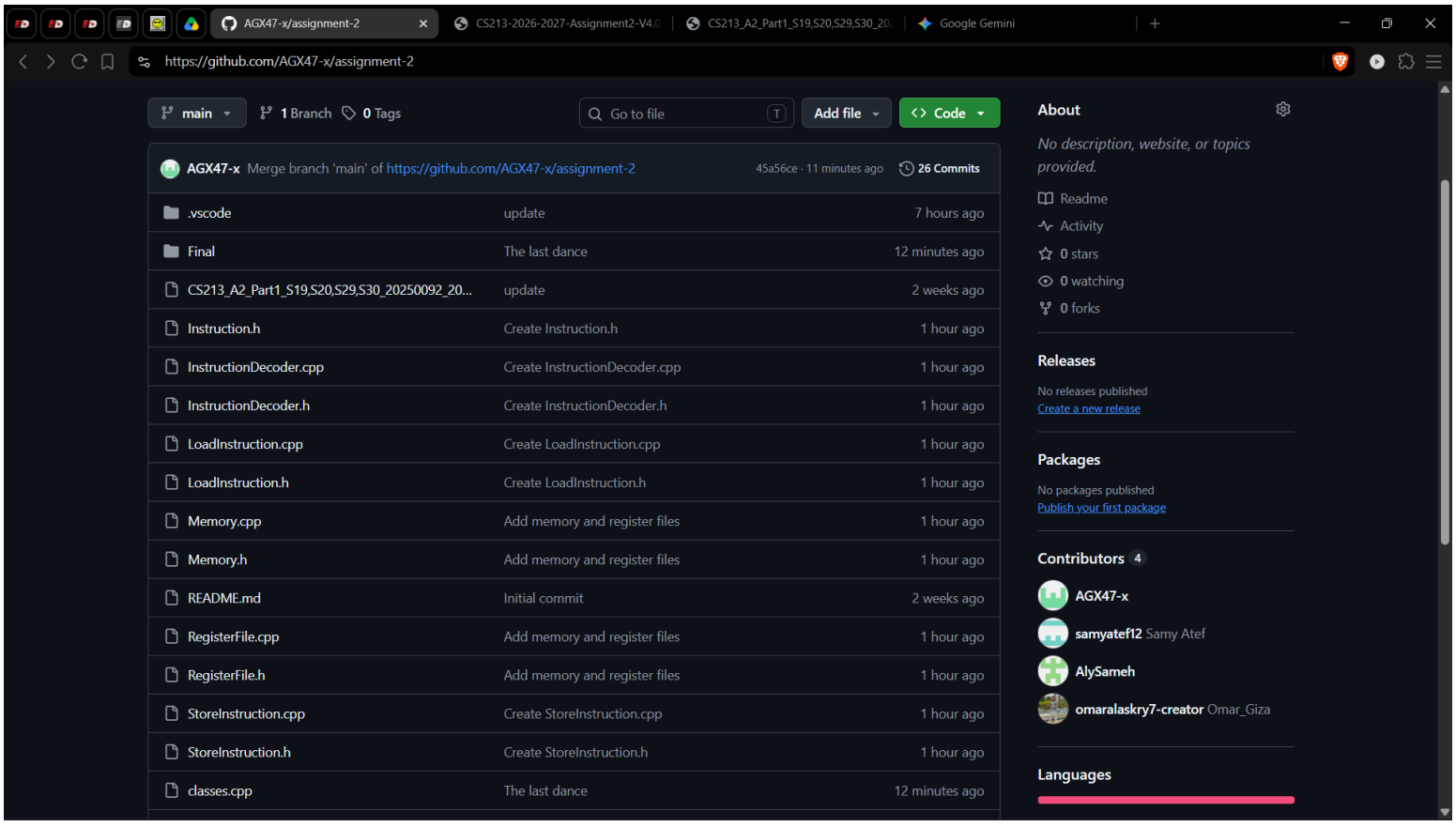
```
210A
220B
230C
5412
5543
260F
8756
37FF
C000
```

; Program 2: Countdown Loop
; This program runs a loop that counts down until it reaches zero.

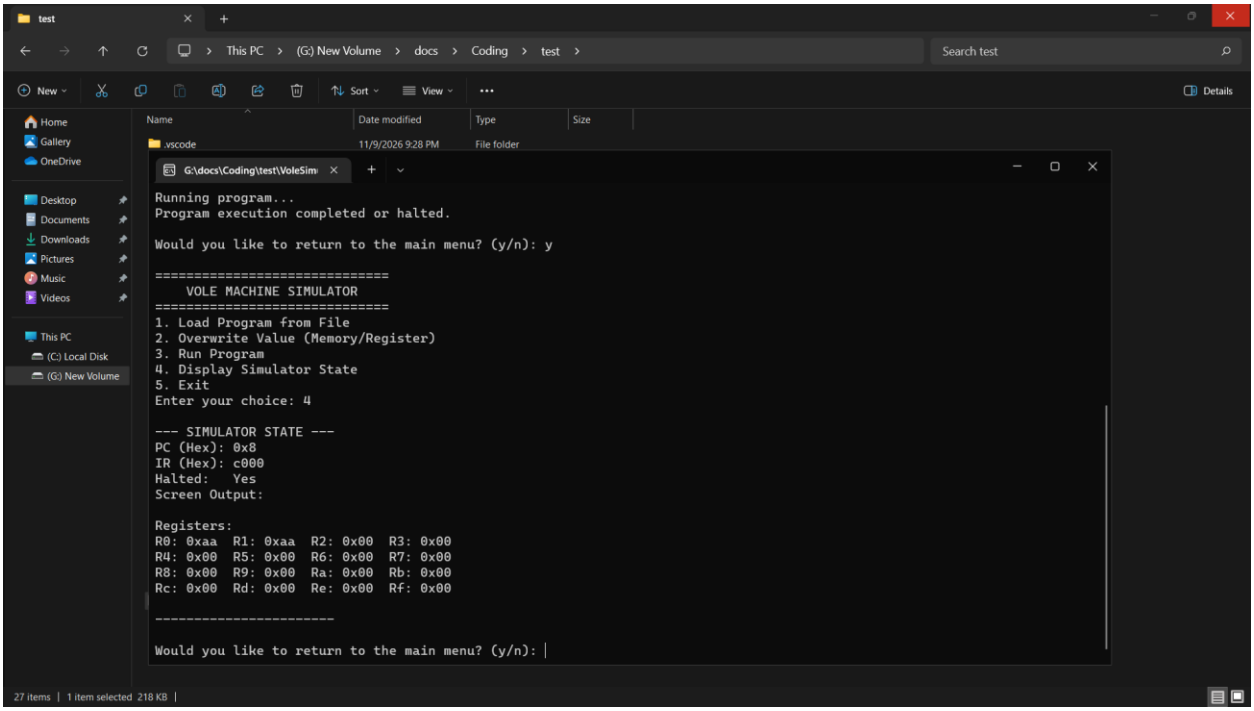
```
2000
2103
22FF
31FF
5112
B10E
B006
C000
```

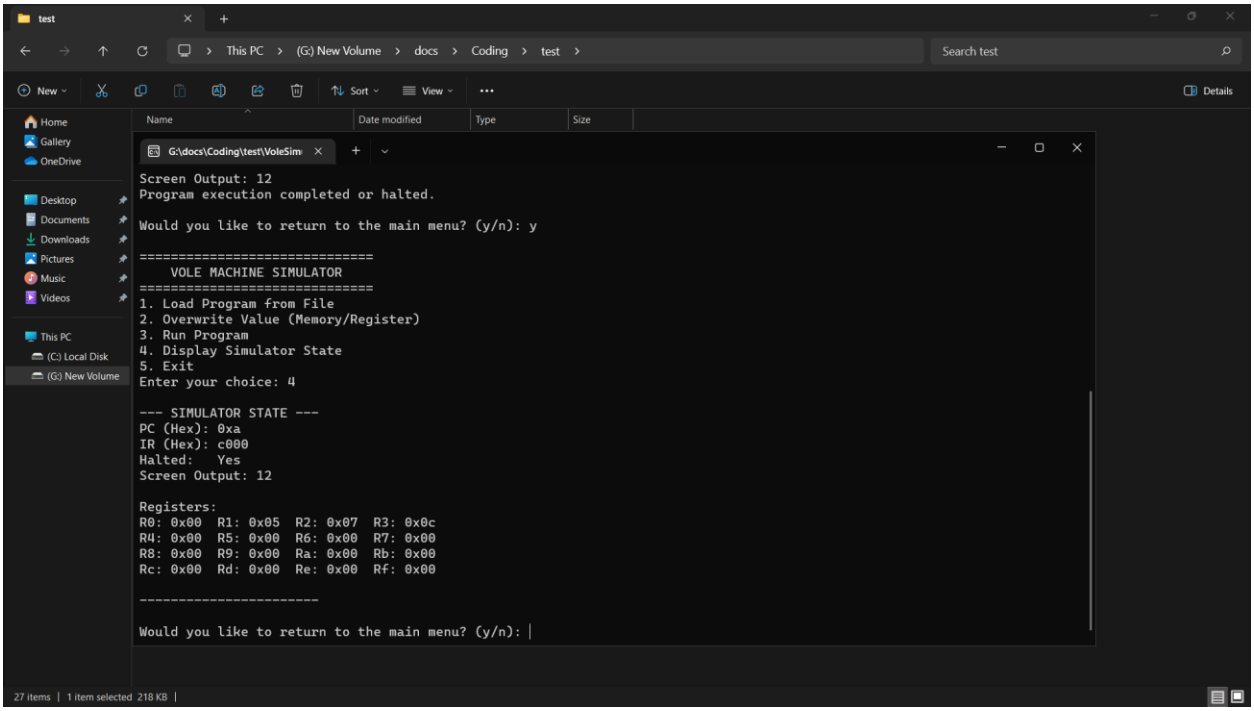
Github

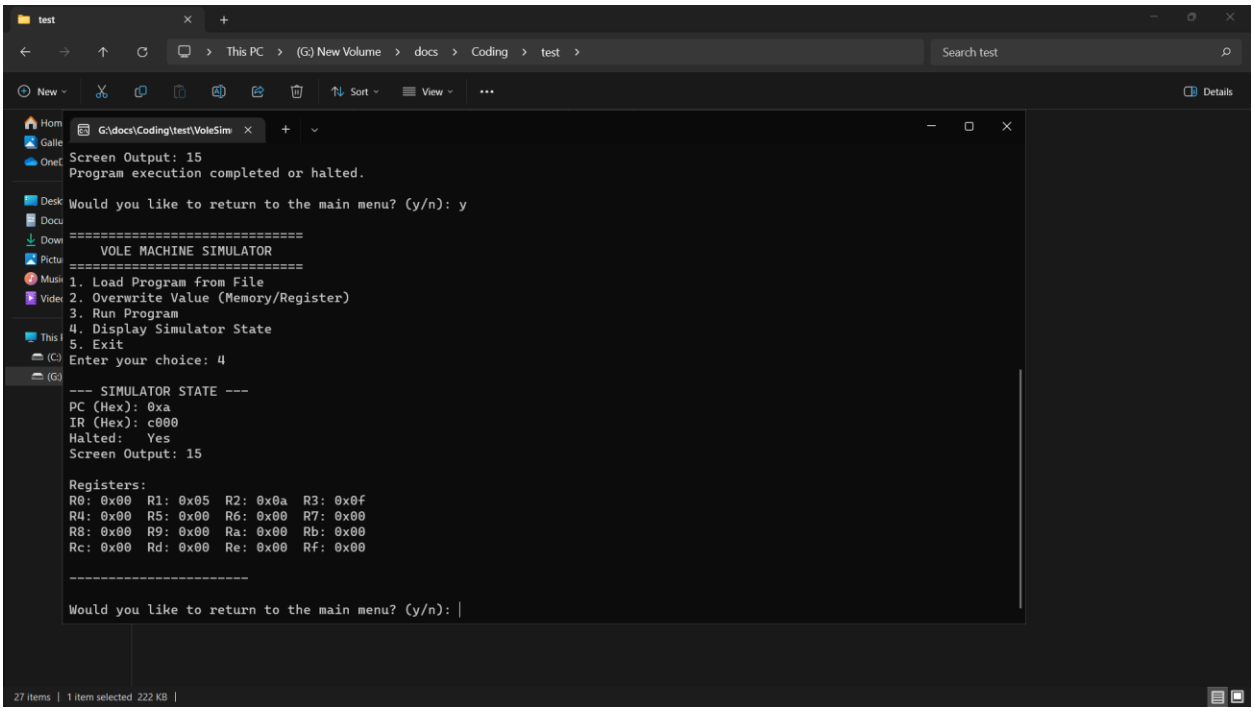

Commits · AGX47-x/assignment-2
https://github.com/AGX47-x/assignment-2/commits/main/
Merge branch 'main' of https://github.com/AGX47-x/assignment-2
AGX47-x committed 9 minutes ago
45a56ce
The last dance
AGX47-x committed 10 minutes ago
f33bce8
Update classes.cpp
samyatef12 authored 1 hour ago
Verified
97cb1da
Create InstructionDecoder.cpp
samyatef12 authored 1 hour ago
Verified
9b22b05
Create InstructionDecoder.h
samyatef12 authored 1 hour ago
Verified
3767566
Create StoreInstruction.cpp
samyatef12 authored 1 hour ago
Verified
bcd5676
Create StoreInstruction.h
samyatef12 authored 1 hour ago
Verified
3531388
Create LoadInstruction.cpp
samyatef12 authored 1 hour ago
Verified
783a980
Create LoadInstruction.h
samyatef12 authored 1 hour ago
Verified
872ae08
Create Instruction.h
samyatef12 authored 1 hour ago
Verified
96cc051
Add memory and register files
omaralaskry7-creator authored 1 hour ago
Verified
dafb29c
Update classes.cpp
samyatef12 authored 2 hours ago
Verified
4294231


AGX47-x/assignment-2
https://github.com/AGX47-x/assignment-2
main
1 Branch
0 Tags
Go to file
Add file
Code
AGX47-x Merge branch 'main' of https://github.com/AGX47-x/assignment-2
45a56ce · 11 minutes ago
26 Commits
.vscode update 7 hours ago
Final The last dance 12 minutes ago
CS213_A2_Part1_S19,S20,S29,S30_20250092_20... update 2 weeks ago
Instruction.h Create Instruction.h 1 hour ago
InstructionDecoder.cpp Create InstructionDecoder.cpp 1 hour ago
InstructionDecoder.h Create InstructionDecoder.h 1 hour ago
LoadInstruction.cpp Create LoadInstruction.cpp 1 hour ago
LoadInstruction.h Create LoadInstruction.h 1 hour ago
Memory.cpp Add memory and register files 1 hour ago
Memory.h Add memory and register files 1 hour ago
README.md Initial commit 2 weeks ago
RegisterFile.cpp Add memory and register files 1 hour ago
RegisterFile.h Add memory and register files 1 hour ago
StoreInstruction.cpp Create StoreInstruction.cpp 1 hour ago
StoreInstruction.h Create StoreInstruction.h 1 hour ago
classes.cpp The last dance 12 minutes ago
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 4
AGX47-x
samyatef12 Samy Atef
AlySameh
omaralaskry7-creator Omar_Giza
Languages

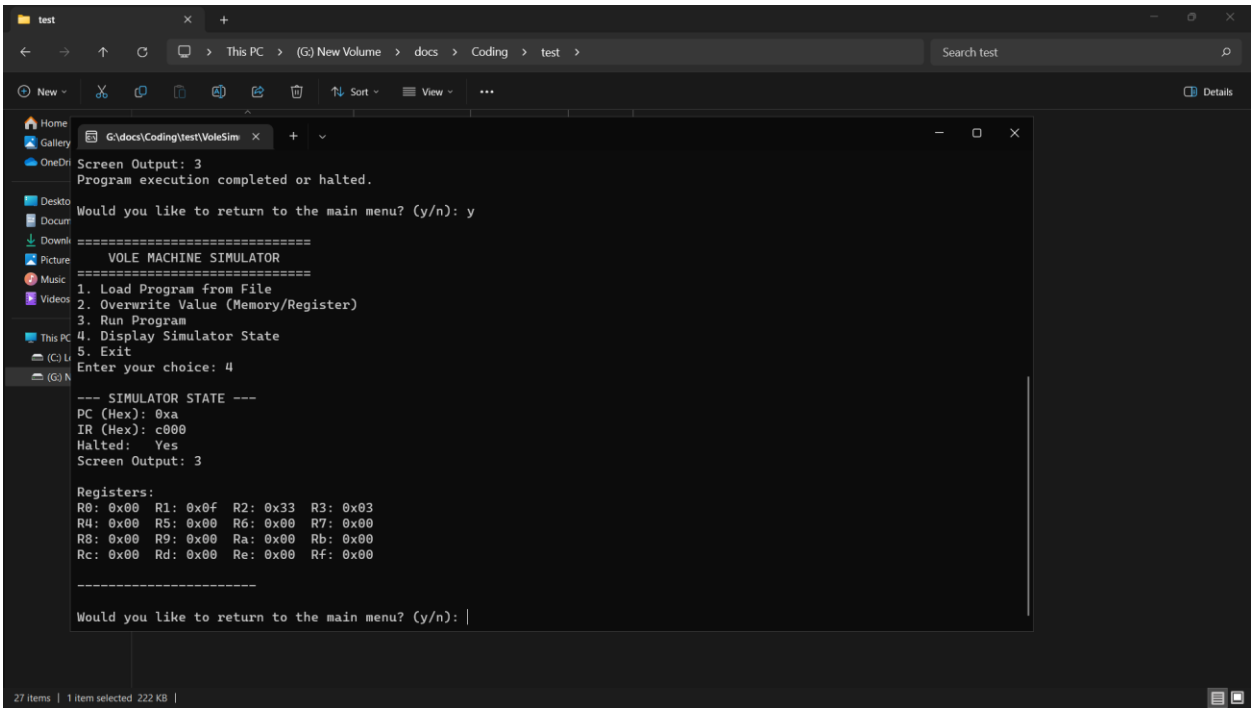
ScreenShots



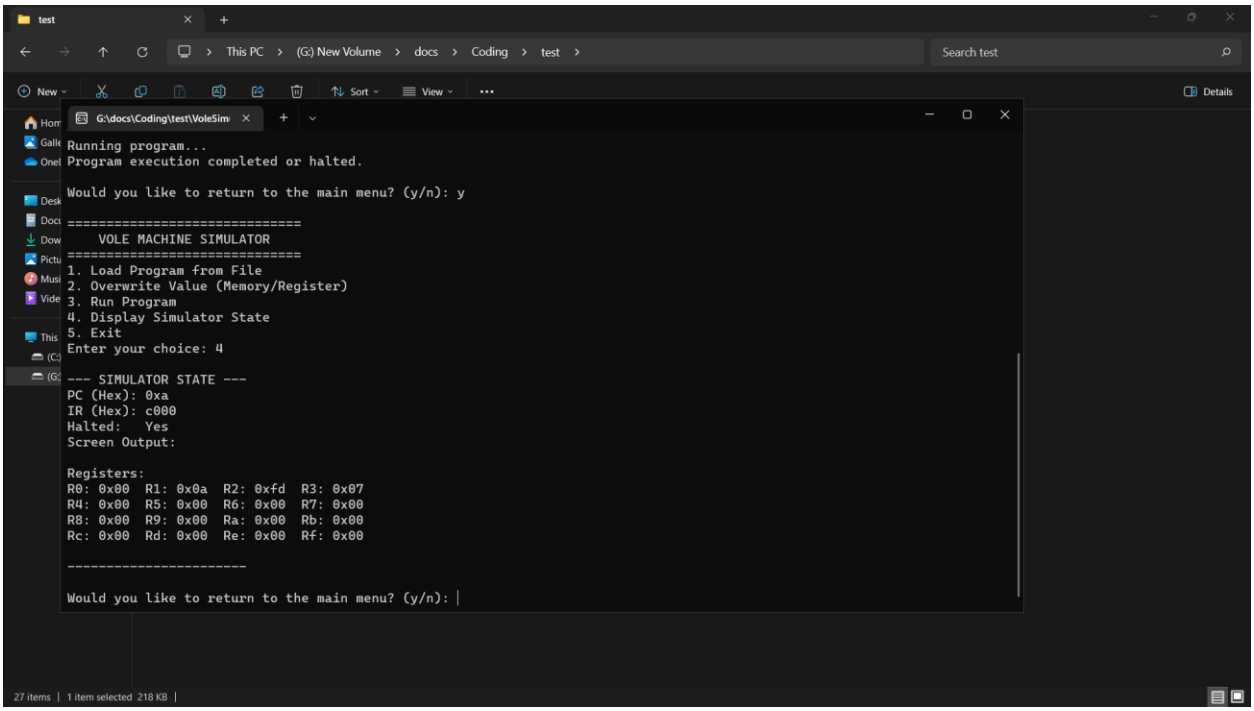



test
This PC › (G:) New Volume › docs › Coding › test ›
Search test
New
Sort
View
Details
G:\docs\Coding\test\VoleSim
Screen Output: 15
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0xa
IR (Hex): c000
Halted: Yes
Screen Output: 15
Registers:
R0: 0x00 R1: 0x05 R2: 0x0a R3: 0x0f
R4: 0x00 R5: 0x00 R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

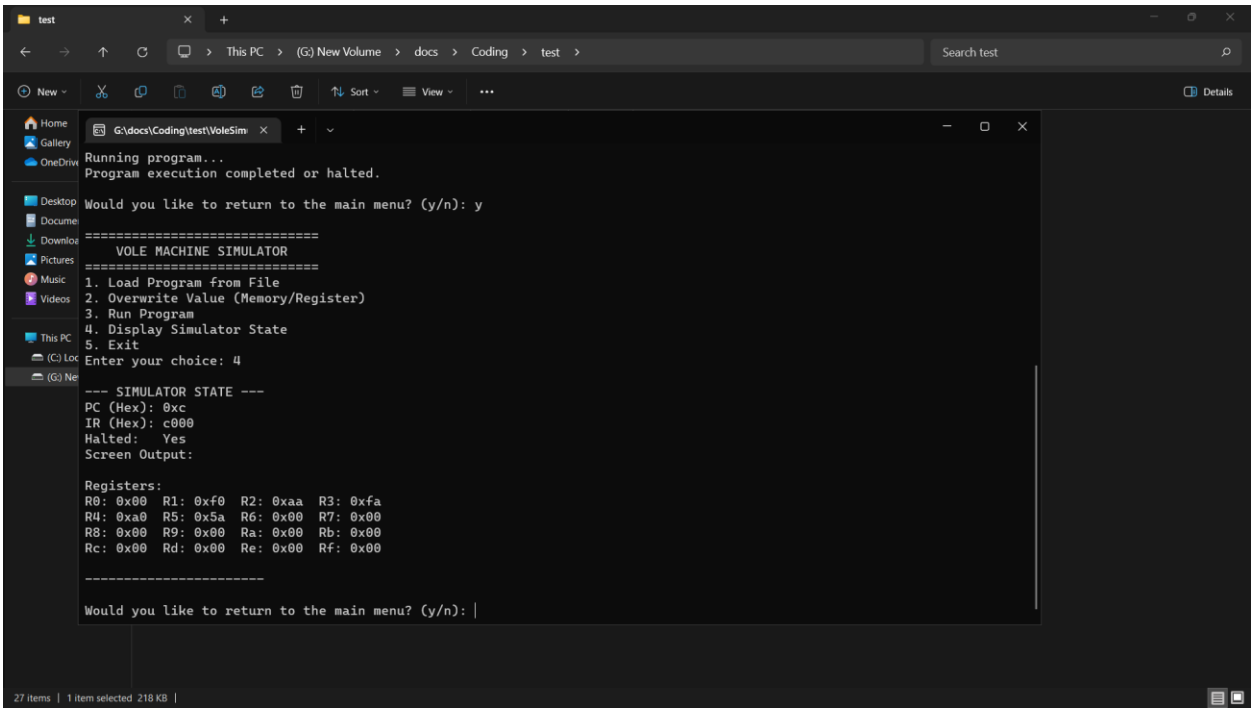
Would you like to return to the main menu? (y/n):
27 items | 1 item selected 222 KB


test
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Screen Output: 3
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0xa
IR (Hex): c000
Halted: Yes
Screen Output: 3
Registers:
R0: 0x00 R1: 0x0f R2: 0x33 R3: 0x03
R4: 0x00 R5: 0x00 R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

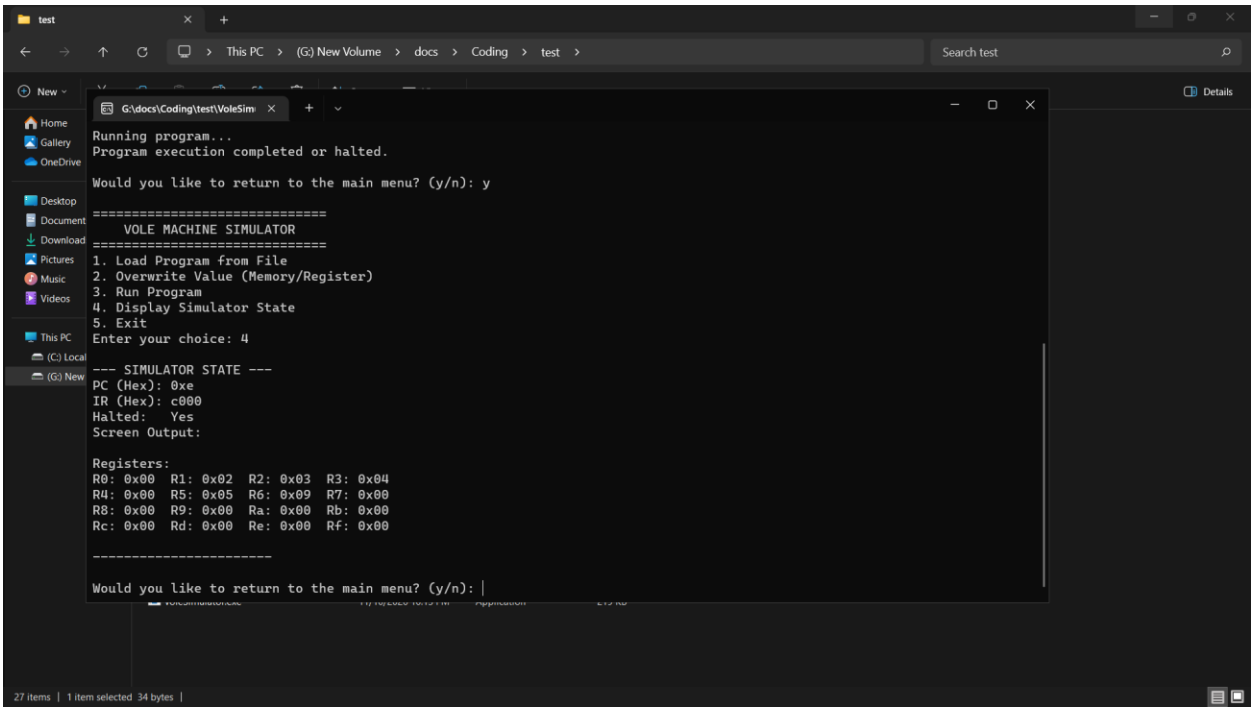
Would you like to return to the main menu? (y/n):
27 items | 1 item selected 222 KB


test
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Search test
New
Sort
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Details
G:\docs\Coding\test\VoleSim
Running program...
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0xa
IR (Hex): c000
Halted: Yes
Screen Output:
Registers:
R0: 0x00 R1: 0x0a R2: 0xfd R3: 0x07
R4: 0x00 R5: 0x00 R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

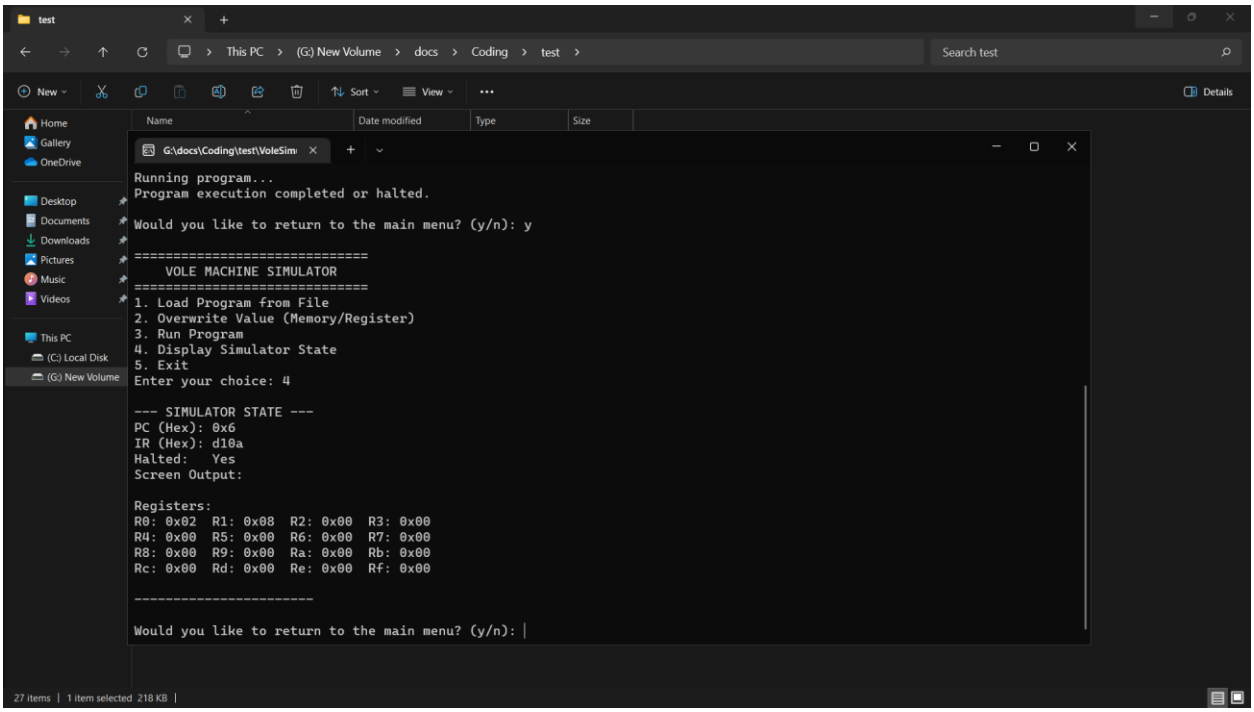
Would you like to return to the main menu? (y/n):
27 items | 1 item selected 218 KB


test
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G:\docs\Coding\test\VoleSim
Running program...
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0xc
IR (Hex): c000
Halted: Yes
Screen Output:
Registers:
R0: 0x00 R1: 0xf0 R2: 0xaa R3: 0xfa
R4: 0xa0 R5: 0x5a R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

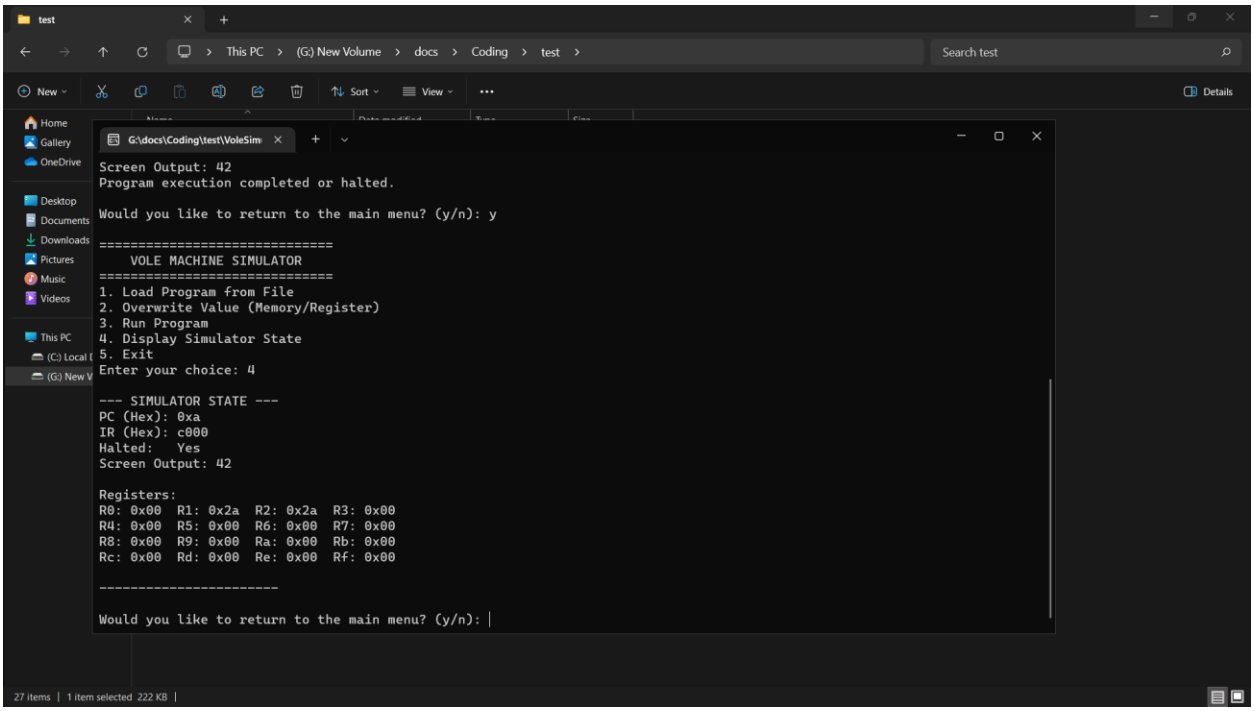
Would you like to return to the main menu? (y/n):
27 items | 1 item selected 218 KB


test
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Search test
New
Details
Home
Gallery
OneDrive
Desktop
Document
Download
Pictures
Music
Videos
This PC
(C:) Local
(G:) New
G:\docs\Coding\test\VoleSim
Running program...
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0xe
IR (Hex): c000
Halted: Yes
Screen Output:
Registers:
R0: 0x00 R1: 0x02 R2: 0x03 R3: 0x04
R4: 0x00 R5: 0x05 R6: 0x09 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

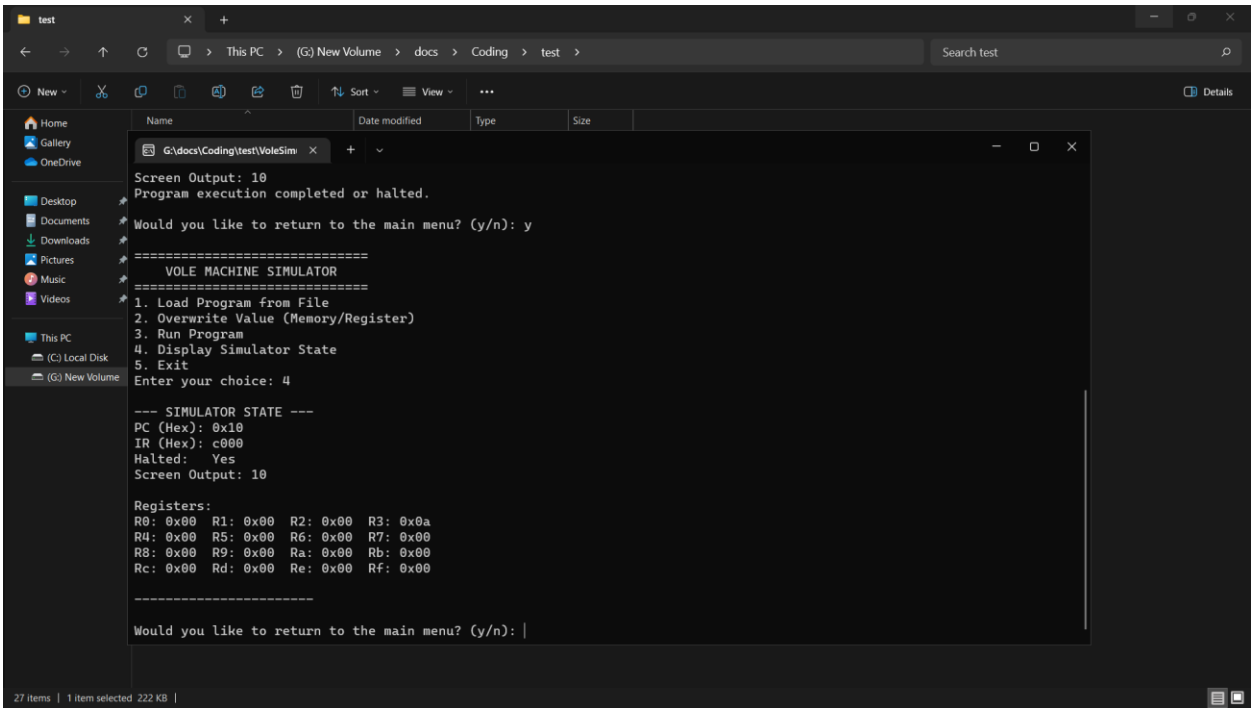
Would you like to return to the main menu? (y/n):
27 items | 1 item selected 34 bytes


test
This PC › (G:) New Volume › docs › Coding › test ›
Search test
New
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View
Details
Home
Gallery
OneDrive
Desktop
Documents
Downloads
Pictures
Music
Videos
This PC
(C:) Local Disk
(G:) New Volume
Name
Date modified
Type
Size
G:\docs\Coding\test\VoleSim
Running program...
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0x6
IR (Hex): d10a
Halted: Yes
Screen Output:
Registers:
R0: 0x02 R1: 0x08 R2: 0x00 R3: 0x00
R4: 0x00 R5: 0x00 R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

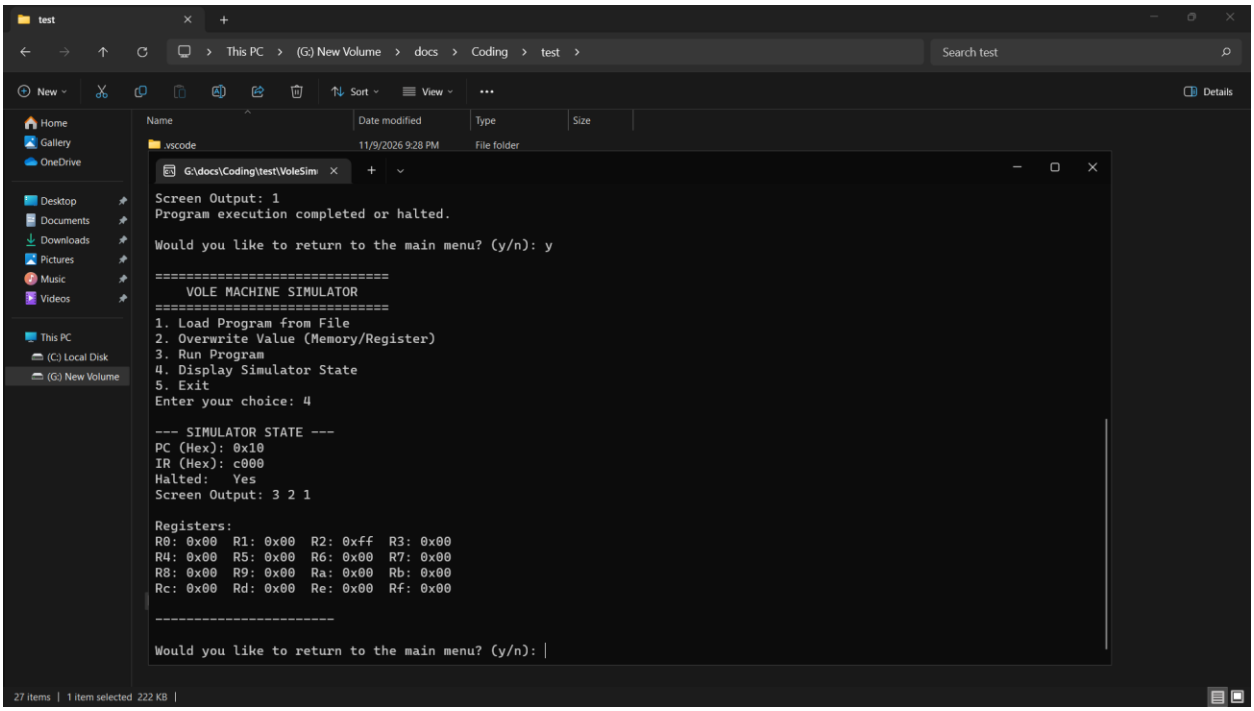
Would you like to return to the main menu? (y/n):
27 items | 1 item selected 218 KB


test
This PC › (G:) New Volume › docs › Coding › test ›
Search test
New
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Home
Gallery
OneDrive
Desktop
Documents
Downloads
Pictures
Music
Videos
This PC
(C:) Local
(G:) New V
G:\docs\Coding\test\VoleSim
Screen Output: 42
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0xa
IR (Hex): c000
Halted: Yes
Screen Output: 42
Registers:
R0: 0x00 R1: 0x2a R2: 0x2a R3: 0x00
R4: 0x00 R5: 0x00 R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

Would you like to return to the main menu? (y/n):
27 items | 1 item selected 222 KB


test
This PC › (G:) New Volume › docs › Coding › test ›
Search test
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Name
Date modified
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Gallery
OneDrive
Desktop
Documents
Downloads
Pictures
Music
Videos
This PC
(C:) Local Disk
(G:) New Volume
G:\docs\Coding\test\VoleSim
Screen Output: 10
Program execution completed or halted.
Would you like to return to the main menu? (y/n): y
==============================
VOLE MACHINE SIMULATOR
==============================
1. Load Program from File
2. Overwrite Value (Memory/Register)
3. Run Program
4. Display Simulator State
5. Exit
Enter your choice: 4
--- SIMULATOR STATE ---
PC (Hex): 0x10
IR (Hex): c000
Halted: Yes
Screen Output: 10
Registers:
R0: 0x00 R1: 0x00 R2: 0x00 R3: 0x0a
R4: 0x00 R5: 0x00 R6: 0x00 R7: 0x00
R8: 0x00 R9: 0x00 Ra: 0x00 Rb: 0x00
Rc: 0x00 Rd: 0x00 Re: 0x00 Rf: 0x00

Would you like to return to the main menu? (y/n):
27 items | 1 item selected 222 KB



Video link:

https://drive.google.com/file/d/1ponEPLA66F3uJvZX4LV4KhWAh7IExR6_/view?usp=drive_link